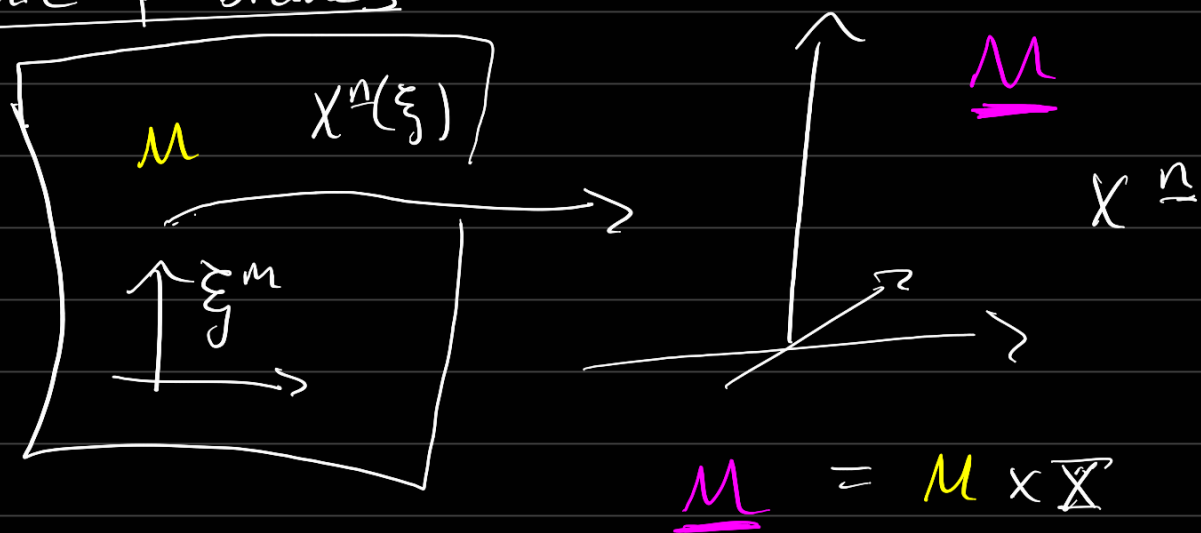


Bosonic p-branes:



$\underline{M}$: Background spacetime	$\dim(\underline{M}) = D$
M : Brane-world volume	$\dim(M) = (p+1)$
$\underline{X}$: perp. coordinates to M	$\dim(\underline{X}) = D-p-1$

$$\xi^m, \quad m=0, \dots, p$$

$$x^{\underline{n}}, \quad \underline{n} = 0, \dots, D-1$$

- Corresponding to tang. space coords $a, b, \dots$ on M and $\underline{a}, \underline{b}, \dots$ on $\underline{M}$. $[T_p] = (\text{mass})^{p+1}$

* point particle: $S_0 = -m \int \sqrt{-g_{\mu\nu} \dot{X}^\mu \dot{X}^\nu} d\tau$

p-brane: $S_p = -T_p \int d\mu_p \mid d\mu_p = \sqrt{-\det(\tilde{g}_{mn})} d^{p+1}\sigma$

$\boxed{\tilde{g} = (X^* g)}$ (pull back) $\tilde{g}_{mn} \equiv g_{\mu\nu}(x) \partial_a X^\mu \partial_b X^\nu$

$\uparrow$ $mn = 0, \dots, p$

for $\underline{p}=0$: point part.
 $\underline{p}=1$: string

G on M
 g on $\underline{M}$

lets look at

(i) $\underline{p}=0$: $\alpha, \beta = 0$ $\tilde{g}_{\alpha\beta} = g_{\mu\nu}(x) \partial_\alpha X^\mu \partial_\beta X^\nu$

$$\tau_0 = m$$

$$\det(\tilde{g}_{00}) = \tilde{g}_{00} = g_{\mu\nu}(x) \partial_\tau x^\mu \partial_\tau x^\nu \quad \checkmark$$

$$\text{ii) } p = 1 \text{ (string)}$$

$$\tau_1 = \frac{1}{2\pi\alpha'}$$

$$\tilde{g}_{mn} = g_{\mu\nu} \partial_m x^\mu \partial_n x^\nu, \quad m, n \in \{0, 1\}$$

$$\det(\tilde{g}_{mn}) = \begin{vmatrix} g_{\mu\nu} \partial_\tau x^\mu \partial_\tau x^\nu & g_{\mu\nu} \partial_\tau x^\mu \partial_\sigma x^\nu \\ g_{\mu\nu} \partial_\sigma x^\mu \partial_\tau x^\nu & g_{\mu\nu} \partial_\sigma x^\mu \partial_\sigma x^\nu \end{vmatrix}$$

$$= (g_{\mu\nu} \partial_\tau x^\mu \partial_\tau x^\nu) (g_{\mu\nu} \partial_\sigma x^\mu \partial_\sigma x^\nu) - (g_{\mu\nu} \partial_\tau x^\mu \partial_\sigma x^\nu) (g_{\mu\nu} \partial_\sigma x^\mu \partial_\tau x^\nu)$$

$$* S_p = -T_p \int d^{p+1} d\mu p$$

-D Invariant under

ii) Diffeomorphisms (of both M & $\underline{M}$)

$$\text{On } \underline{M}: \quad x^\mu \rightarrow x'^\mu(x^\mu), \quad \xi^\mu \rightarrow \xi'^\mu \text{ (inert)}$$

$$x'^\mu = \frac{\partial x'^\mu}{\partial x^\mu} x^\mu$$

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}(x)$$

$$\partial_n x^\mu = \frac{\partial x^\mu}{\partial \xi^n} \rightarrow \partial_n x'^\mu = \partial_n \left(\frac{\partial x'^\mu}{\partial x^\mu} x^\mu \right)$$

$$\text{How does } \partial_n \left(\frac{\partial x'}{\partial x} \right) = 0?$$

$$\text{On } M: \quad \xi^\mu \rightarrow \xi'^\mu(\xi), \quad x'^\mu(\xi') = x^\mu(\xi)$$

and so

$$g'_{\mu\nu}(x') = g_{\mu\nu}(x)$$

$$\partial_n x^\mu \rightarrow \partial_{n'} x^\mu = \frac{\partial x^\mu}{\partial \xi^{n'}} \partial_{n'} x^\mu$$

$$\rightarrow \det(g'_{mn}) = J^2 \det(g_{mn})$$

$$J = \det \left(\frac{\partial \xi^\mu}{\partial \xi'^\mu} \right)$$

$$\text{but } d^{p+1} \xi' = J^{-1} d^{p+1} \xi$$

* Brane does not have enough symmetry to determine its couplings to the fields in the background S.T. other than gravity.

→ Brane couples to a rank $(p+1)$ -Gauge field $A_{\underline{m}_1, \dots, \underline{m}_{p+1}}(x^\mu)$ living on the background spacetime via

$$q \int d^{p+1} \xi \partial_{\underline{n}_1} x^{\underline{m}_1} \dots \partial_{\underline{n}_{p+1}} x^{\underline{m}_{p+1}} A_{\underline{m}_1 \dots \underline{m}_{p+1}} \epsilon^{\underline{n}_1 \dots \underline{n}_{p+1}}$$

q : charge of the brane

* Action is invariant

• On M : $x^\mu \rightarrow x'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} x^\nu$

$A_{\underline{m}_1 \dots \underline{m}_{p+1}} = \frac{\partial x^{\underline{p}_1}}{\partial x'^{\underline{m}_1}} \dots A_{\underline{p}_1 \dots \underline{p}_{p+1}}$

• On M : $\xi^m \rightarrow \xi'^m$

$$J \epsilon^{\underline{n}_1 \dots \underline{n}_{p+1}} = \left(\frac{\partial \xi^{\underline{n}_1}}{\partial \xi'^{\underline{q}_1}} \dots \right) \epsilon'^{\underline{q}_1 \dots \underline{q}_{p+1}} = J \epsilon^{\underline{n}_1 \dots \underline{n}_{p+1}}$$

ex: D-brane: $x^\mu(\tau)$, $\xi^0 = \tau$

$$S_0 = q \int d\tau \frac{dx^\mu(\tau)}{d\tau} A_\mu(\tau) \quad | \quad A = A_\mu(\tau) dx^\mu$$

$$= q \int A$$

$$\tilde{S}_0^{\text{tot}} = -m \int d\tau \sqrt{-g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu} + q \int d\tau \frac{dx^\lambda}{d\tau} A_\lambda$$

* E.O.M of x^μ :

$$\partial_\tau \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = -m \left(-g_{\mu\nu} \dot{x}^\nu \right) + q \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} A_\lambda$$

$$\frac{d}{d\tau} \left(\frac{\dot{x}^\mu}{\sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}} \right) + \dots$$

$$m \frac{d}{d\tau} \left(\frac{\dot{x}_P}{\sqrt{-x^2}} \right) + q \frac{dx^q}{d\tau} \partial_q A_P$$

$$= m \frac{d}{d\tau} \left(\frac{\dot{x}_P}{\sqrt{-x^2}} \right) + q \frac{dx^q}{d\tau} \partial_q A_P$$

$$\frac{\partial \mathcal{L}}{\partial x^\mu} = -m \frac{(\partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta}{\sqrt{-x^2}} \quad \left(\text{term vanishes} \right)$$

(P_μg = 0)

$$+ q \frac{dx^q}{d\tau} \partial_P A_q$$

∴ E.O.M:

$$m \frac{d}{d\tau} \left(\frac{\dot{x}_P}{\sqrt{-x^2}} \right) + q \frac{dx^q}{d\tau} F_{qP} = 0$$

• Gauge choice $\tau = t = x^0$

$$F_{i0} = E_i, \quad F_{ij} = \epsilon_{ijk} B_k, \quad i, j = 1, \dots, D-1$$

• Non-rel. limit: $-\dot{x}^2 = 1$

$$m \ddot{x}_P = -q \frac{dx^0}{d\tau} F_{0P} - q \frac{dx^i}{d\tau} F_{iP}$$

$$= +q E_i \delta_{iP} +$$

$$\ddot{x}_0 = 0$$

$$\ddot{x}_i = +q E_i + q \frac{dx^j}{d\tau} F_{ji}$$

$$= +q E_i + q \frac{dx^j}{d\tau} \epsilon_{ijk} B_k$$

$$\ddot{x}_i = q \left(E_i + \epsilon_{ijk} \frac{dx^j}{d\tau} B_k \right)$$

$$\underline{\ddot{x}} = \ddot{x}_i dx^i, \quad \underline{E} = E_i dx^i, \quad \text{etc...}$$

$$\underline{\ddot{x}} = q \left(\underline{E} + \frac{d\underline{x}}{d\tau} \wedge \underline{B} \right)$$

West has $\wedge$ But this
would be a 2-form...

A 1-brane (string) couples to a two form

$A_{m_1 m_2}$

$$q \int d^2 \xi \epsilon^{mn} \partial_m X^{m_1} \partial_n X^{m_2} A_{m_1 m_2}$$

* In general, the above should be invariant under a gauge transformation:

$$\delta A_{\underline{m}_1 \dots \underline{m}_{p+1}} = (p+1) \partial_{\underline{m}_1} \Lambda_{\underline{m}_2 \dots \underline{m}_{p+1}}$$

→

$$g(p+1) \int d^{p+1} \xi e^{n_1 \dots n_{p+1}} (\partial_{\underline{m}_p} x^{\underline{m}_1} \dots) \partial_{\underline{m}_1} A_{\underline{m}_2 \dots \underline{m}_{p+1}}$$

$$= -g(p+1) \int d^{p+1} \xi e^{n_1 \dots n_{p+1}} x^{\underline{m}_1} \partial_{\underline{n}_2} x^{\underline{m}_2} \partial_{\underline{m}_1} \partial_{\underline{n}_1} A_{\underline{m}_2 \dots \underline{m}_{p+1}}$$

$$= g(p+1) \int d^{p+1} \xi e^{n_1 \dots n_{p+1}} (\partial_{\underline{n}_2} x^{\underline{m}_2} \dots) \partial_{\underline{n}_1} A_{\underline{m}_2 \dots \underline{m}_{p+1}}$$

$$= g(p+1) \int d^{p+1} \xi e^{n_1 \dots n_{p+1}} \partial_{\underline{n}_1} (\dots) \Lambda_{\underline{m}_2 \dots \underline{m}_{p+1}}$$

= 0

so long as $\Lambda_{\underline{m}_2 \dots \underline{m}_{p+1}}|_{\partial M} = 0$
 note the

$\partial_{\underline{n}_1} \partial_{\underline{n}_i}$ terms vanish
 since $\epsilon^{n_1 n_i} \partial_{\underline{n}_1} \partial_{\underline{n}_i}$

• Break $\underline{n}, \underline{m}$ into

$$\underline{n} = (n, n')$$

$$\underline{n} : 0, 1, \dots, D-1$$

$$n : 0, 1, \dots, p$$

$$n' : p+1, \dots, D-1$$

• As in Part. Physics we can find a way to reduce the # of D.O.F via a gauge choice

→ Static Gauge: Use the dotted invariance of the world-sheet

Identify the coordinates on D_p with those of the embedding maps: i.e. its an inclusion

$$D_p \hookrightarrow \underline{M}$$

$$\sqrt{-g} = 1 - \sum n$$

$$X(\xi) = g, \quad n = 0, 1, \dots, p$$

$\rightarrow$ leaves $X^{n'}(\xi)$, $n' = p+1, \dots, D-1$ to describe the dynamics of the brane $(D-p-1)$ -coords

* Think of $(D-p-1)$ as the Goldstone Bosons or, Zero modes, of the broken translation due to the presence of the p -brane

$\rightarrow$ Symmetry breaking: $G \rightarrow H$
 $|G/H|$: # of G.S. Bosons

• Do a QFT on the worldvolume of the brane which has Poincare invariance.

$$\underline{M} \triangleq ISO(1, D-1)$$

Broken
 $\downarrow$

$$D_p \otimes S^{D-p}: ISO(1, p) \times SO(D-p-1)$$

$$\therefore \left| \frac{ISO(1, D-1)}{ISO(1, p) \times SO(D-p-1)} \right| = \# \text{ of Bosons}$$

$$SO(N): O^T O \quad \det(O) = 1$$

$$e^{x^T} e^x = 1$$

$$(1 + x^T)(1 + x) = 1$$

$$x^T x + x x^T = 0 \Rightarrow x^T x = -x x^T$$

$$\det(e^x) = e^{\text{tr}(x)} = 1$$

$$1 + \text{tr}(x) = 1$$

$$\text{tr}(x) = 0$$

$$\Rightarrow \sum_{i=1}^n \frac{n^2 - n}{2} = \frac{n(n-1)}{2}$$

$$\dim(\mathfrak{so}(N)) = \frac{N(N-1)}{2}$$

$$\dim(\mathfrak{ISO}(1, D-1)) = \mathbb{R}^{1, D-1} \ltimes \mathfrak{O}^\uparrow(1, D-1)$$

$$= \frac{D + D(D-1)/2}{D(D+1)/2}$$

$$\left| \frac{\mathfrak{ISO}(1, D-1)}{\mathfrak{ISO}(1, p) \times \mathfrak{SO}(D-p-1)} \right| = \frac{D^2 + D}{2} - \frac{(p+1)^2 + (p+1)}{2} - \frac{(D-p-1)(D-p-2)}{2}$$

$$= D - p - 1 \quad ? ? ? ?$$

ex: M2 & M5 branes in 11D:

$$\underline{M2}: 11 - 3 = 8 : \text{Bosons}$$

$$\underline{M5}: 11 - 6 = 5 : \text{Bosons}$$

→ Only the translational symmetry is broken

Note that the Gauge action is simply

$$\int A_{(p+1)}$$

as a top form.

* Solutions of p-branes only depended on harmonic fns N which depends on the $D-p-1$ transverse coords.

Monopole: (0-brane) In $D=4$ (Soliton)

→ Dirac is a static solution in $N=2$ susy Y.M.

→ Moduli space: space of parameters required to specify the solution

• For K monopoles $|M| = 4K$

$K=1$: $M \cong \mathbb{R}^3 \otimes S^1 \xrightarrow{\text{pushout}}$ "B"-term: gauge transf.

→ $\exists$ a static solution: i.e. they experience no forces while at rest.

→ For low energies just above their rest mass we can

approx. motion in terms of their moduli space
 $\hookrightarrow$ Motion is described by an action whose dynamical D.O.F. / variables are coordinates of the moduli space

$K=1$: $X^i(\tau)$, $i=1,2,3$, $\mathcal{M}(\tau)$ on S^1
 $\xi^0 \equiv \tau$ (see next)

- Broken translational symmetry: non-linear realisation which includes Goldstone bosons

$\hookrightarrow$ (Moduli Parameters)
 X^i , $i=1,2,3$

* Consider the Moduli-space and Action which describes the low energy behavior of p-brane solitons.

Types of Superbranes: (Super p-brane)

* Can be seen as a $(p+1)$ -dim Bosonic submfld M w/ ξ^n , $n=0, \dots, p$ that moves through a background superspace $\underline{M}$ w/ coords

$$Z^M = (X^A, \Theta^\alpha), \quad \Theta^\alpha: \text{Spinor}$$

$\underline{M}, \underline{N}$: World indices (AdS) |

$\underline{A}, \underline{B}$: tangent indices (Lorentz) |

- Simple superbranes: $Z^M(\xi^n)$ described entirely by specifying the Superworld surface the p-brane sweeps out

$\hookrightarrow$ (Type I branes)

- Also have branes which have fields living on their world volume.

$$S = S_1 + S_2 \quad \left| \quad \begin{aligned} S_1 &= -T \int d^{p+1} \xi \sqrt{-\det(g_{mn})} + \dots \\ S_2 &= q \int A_{(p+1)} + \dots \end{aligned} \right.$$

$\xi^m \quad \xi^n$

where $\begin{cases} g_{mn} = \partial_n Z^M \partial_m Z^M g_{MN} \\ g_{MN} = E_N^a E_M^b \eta_{ab} \end{cases} \quad Z^M = (X^A, \Theta^\alpha)$

E_N^a : Super vielbein on the background superspace

E_N : supervising on the background super space.
 • $+ \dots$: denotes terms involving fields living on the brane world volume and background fields

$$[T] = (\text{mass})^{p+1}$$

- For a static brane $T = \left(\frac{\text{mass}}{\text{Vol}} \right)$ on the brane
- g_{NM} : not a background metric in the conventional sense when

$$g_{NM} = \underbrace{E_N^a E_M^b \eta_{ab}}_{\substack{\text{only runs over} \\ \text{bosonic coords}}}$$

* Spacetime vs brane world Susy:

→ there our Superbranes possess space-time Susy which means the background fields must belong to supermultiplets of the background superspace.

→ Background fields are specified by the background Susy of the brane for 32 Supercharges.

- For 16 Supercharges: larger choice of background supergrav. Mott.

$S = S_1 + S_2$: invariant under super-reparametrizations of the Background spacetime by definition. M

- Only invariant w.r. Bosonic of the world surface M

- Also inv. under K_2 -symmetry

→ In flat-Background spacetime $\eta_{NM} = E_N^a E_M^b \eta_{ab}$

$$\underline{N} = (\underline{n}, \underline{\alpha})$$

$$\partial_m \underline{z}^N E_N^a = \partial_m X^{\underline{n}} E_{\underline{n}}^a + E_{\underline{\alpha}}^a \partial_m \Theta^{\underline{\alpha}}$$

$$E_{\underline{n}}^a = e_{\underline{n}}^a$$

$$E_{\underline{\alpha}}^a = -\frac{i}{2} \bar{\Theta}^{\underline{\beta}} (\gamma^a)_{\underline{\beta} \underline{\alpha}} \Theta^{\underline{\alpha}}$$

$$\partial_m \underline{z}^N E_N^{\underline{\alpha}} = \partial_m \Theta^{\underline{\alpha}} \Rightarrow$$

$$\begin{aligned} E_{\underline{n}}^{\underline{\alpha}} &= 0 \\ E_{\underline{\beta}}^{\underline{\alpha}} &= \delta_{\underline{\beta}}^{\underline{\alpha}} \end{aligned}$$

$$\therefore E_{\underline{N}}^{\underline{M}} = \begin{pmatrix} e_{\underline{n}}^{\underline{a}} & 0 \\ 0 & \delta_{\underline{\beta}}^{\underline{\alpha}} \end{pmatrix}$$

$$\left(-\frac{i}{2} \bar{\Theta}^\alpha \gamma^\alpha \right)_{\underline{B}\underline{A}} \quad \delta_{\underline{B}}^{\underline{A}} \quad \right)$$

* Thus in a flat background spacetime

$$g_{mn} = \partial_n Z^N E_N^a \partial_m Z^M E_M^b \eta_{ab}$$

$$g_{mn} = \left(\partial_m x^a - \frac{i}{2} \bar{\Theta} \gamma^a \partial_m \Theta \right) \left(\partial_n x^b - \frac{i}{2} \bar{\Theta} \gamma^b \partial_n \Theta \right) \eta_{ab}$$

$$\Pi_m^a \equiv \partial_m Z^N E_N^a = \partial_m x^a - \frac{i}{2} \bar{\Theta} \gamma^a \partial_m \Theta$$

→ We get the Green-Schwarz action for a superstring
for a super-1-brane: G.S. has spacetime susy

* M-theory: 11-D susy and so M2 and M5 have the
above form with Background given by 11D sugra
multiplet $([2000]_9 \oplus [1001]_4 \oplus [0010]_9)$

↑
[3]₉

* G-S: Background Susy

NSOR: Worldvolume susy

Take G-S to light cone coordinates: Hidden world volume
susy.

Take to light cone coordinates shows NSOR has hidden
spacetime (Background) susy (provided we carry out GSO)
projection

Hence:

$$(G.S. \cong NS \otimes R) \quad (\text{same theory})$$

* Does not hold for a point particle.

• branes with K-symmetry in G-S formulation have a hidden
world volume susy.

• NSOR: has a superembedding formalism: sweeps out a supermanifold

• Posses a Super static Gauge for which the Bosonic coordinates
take the form $X^{\underline{a}}(\xi) = \xi^{\underline{a}}, \quad \underline{a} = 0, \dots, p$

→ K-symm can be used to set half the fermions $\Theta^{\underline{A}}(\xi)$
 $= (\Theta^{\underline{\alpha}}(\xi), \Theta^{\underline{\beta}}(\xi))$ to vanish $\Theta^{\underline{\alpha}} = 0$

• Remaining $X^{\underline{a'}}$, $\underline{a'} = p+1, \dots, D-1$: Goldstone Bosons

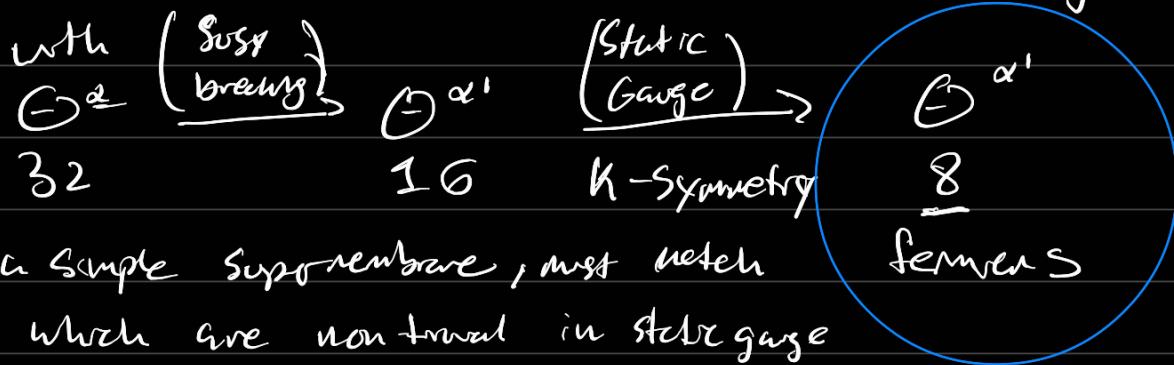
$\ominus^{\alpha'}$: Goldstone fermions (Goldstinos)

$\hookrightarrow$ From half susy broken by the branes

* Since the branes contain world volume susy: fields living on the surface must be part of a supermultiplet
 $\Rightarrow$ Equal # of D.O.F. on-shell

(i) Consider a theory w/ a p-brane that has maximal background susy: brane moves in a background superspace corresponding to maximal supergravity theory

• Brane breaks half the 32 supercharges, left



* For a simple supermembrane, must match $X^{\mu'}$ which are non-trivial in static gauge

$$X^{\mu'} = (p+1), \dots, D-1$$

$$D - p - 1 = 8$$

$$\text{or } D - 8 = (p+1)$$

$$D - (p+1)$$

$$= D - p - 1$$

(Simple Supermembranes) $\left(\begin{smallmatrix} \ominus^{\alpha'} \\ \underline{8} \end{smallmatrix} \oplus \begin{smallmatrix} X^{\mu'} \\ \underline{8} \end{smallmatrix} \right)$

Chiral mult.

D=11: p=2: M2-brane is the only simple superbrane with maximal background susy.

D=10: p=1: (1-brane) / string with $B(2)$

$Q: R \otimes R$ or $NS \otimes NS$ brane?

* Super D-branes: (p-branes) $\uparrow$

• Requires a gauge (vector) field A_n , $n=0, \dots, p$ living in the brane world volume in addition to $(X^{\mu}, \ominus^{\alpha})$

$\rightarrow$ Think this answers this: the $R \otimes R$ sector, which has Dirichlet B.C.'s for the string.

$\ominus^{\alpha'}$ (fermions)	}	$X^{\mu'} \oplus A_n, n=0, \dots, p$
<u>8</u>		Bosons $\downarrow$

$$D-p-1 \quad (p-1)$$

• A_n : has $(p-1)$ D.O.F. (on-shell) due to Gauge fixing.

$$D-p-1 + p-1 = 8 \Rightarrow \boxed{D=10}$$

→ Can only have maximal D-branes in $D=10$

• Super p-brane: couples to a $(p+1)$ -gauge field

Belongs to background sugra.

$$\begin{array}{l|l} A_{(p+1)} \oplus B_{(D-p-3)} & F = dA \quad G = *F \quad d*F = 0 \\ [p+1] \oplus [D-p-3] & C = dB \quad dG = 0 \end{array}$$

$$G_{n_1 \dots n_{D-p-2}} = \frac{1}{(p+2)!} G_{n_1 \dots n_{D-p-2} \dots n_D} F_{n_{D-p-1} \dots n_D}$$

$$\underline{11D}: M_2 \oplus M_5$$

$$A^{(3)} \quad B^{(5)}$$

• M2: Simple super brane: 8 Goldstone modes $X^{n'}$, $n' = 1, \dots, 3$

M5: 5 Goldstone modes $X^{n'}$, $n' = 1, \dots, 6$

→ Goldstone fermions: from susy breaking + R-sym

$$32 \rightarrow 16 \xrightarrow{\text{on-shell}} \underline{8}$$

* Thus, we need an additional 3 Bosonic D.O.F. which form massless reps of $SO(4)$ of the worldvolume

$$\underline{3} \text{ of } SO(4) \cong SU(2) \times SU(2) \quad \underline{3} = [2, 0]_4 \quad (\text{Self-dual 2-form})$$

$$\text{Diagram: } \begin{array}{c} \text{Red circle} \quad \text{Blue circle} \quad D_n \quad \text{Red circle} \quad \text{Blue circle} \\ \text{O} - \text{O} - \text{O} \end{array}$$

Q: There also exists an antiseif dual rep of $SO(4)$ why is that not an option?

A: 11D has only one Maj. spinor which is not Maj-Weyl

So we can only build $A^{\alpha\beta} (\gamma_{\mu\nu})_{\alpha\beta} \rightarrow A_{\mu\nu}$

• $\gamma_{\mu\nu}$: Generators of $Spin(2)$

$$A^{(2)}: \quad *A^{(2)} = A^{(2)}$$

↑?

$$\text{on-brane: } 6 \quad D-2: 4$$

$$\text{space-like: } 5 \quad D-2: 3$$

So, how is this self dual? what geometry is * with respect to?

• Also, where does this 2-form come from in terms of

the supergravity mult $(\psi_\mu, \tilde{e}^\mu, e_N)$:

10D:

IIA: $[1000]_8, [0100]_8, [0010]_8$
 $[0100]_8$: NS - 2 form $(NS \otimes NS) \rightarrow [6]_8$ $[6]_8$
 $([1000]_8 \oplus [0010]_8) : (R \otimes R)$ D-branes
 + D-branes
 $(NS \otimes NS) [2]_8 \oplus [6]_8 (F1 \oplus NS5)$ branes
 $(R \otimes R) [1]_8, [3]_8, [5]_8, [7]_8$
 D_0, D_2, D_4, D_6 branes

Aside: For $D=11, 12$: $\frac{8}{8}$ fermionic Goldstone
 $X^{n+} = 11, \dots, 3$ $\frac{8}{8}$ Goldstone Bosons
 $\rightarrow$ so because we don't need extra terms this makes the brane simple

* For type IIA: $F1$ is the only simple brane
 $D = p-1 = 8$, $D=10$: $10-9 = p = 1$
 $\rightarrow F1$ couples to $A_{\mu\nu}^{(2)}$ (Type IIA string)
 $\rightarrow NS5_A \rightarrow$ dim reduced brane from 11D

If D-branes have gauge field living on their world volume
 $\rightarrow$ 8-brane which is non dynamical

$\hookrightarrow \int A^{(9)}$ Gauge field
 $\left[A^{(1)} \right]$
 $F^{(10)} : -\frac{1}{6!} \int d^{10}x \sqrt{-\det(g_{\mu\nu})} F^{n_1 \dots n_{10}} F_{n_1 \dots n_{10}} \Big|_{F_{10} = C \in_{10}}$
 $: \int d^{10}x \frac{c^2}{\sqrt{-\det g_{\mu\nu}}} \quad (\text{cosmological constant})$

II B: $[0100]_8 \in NS \otimes NS (F1 \oplus NS5) A^{(6)}$
 $[0000]_8 \oplus [0100]_8 \oplus [0020]_8 \in R \otimes R$
 $\rightarrow [0] \oplus [2] \oplus [4] \oplus [6] \oplus [8]$
 $-2, 1, 3, 5, 7$ branes
 D_1, D_3, D_5, D_7
 $\uparrow$ Instanton

D_3 : couples to the self dual $A^{(4)}$ and has self duality symmetry

$\rightarrow$ has two 1-branes: $1_s + 1_D$

M theory								S: Simple
IIA	0_D	1_s	2_D	4_D	5	6_D	8_D	D: D-brane
IIB		$1_s + 1_D$	3_D		$5 + 5_D$	7_D		$\uparrow$ R & R

* excludes D_{-1} instanton

• p-brane actions can be thought of as a low energy action of these solitons

• Current:
$$j^{\mu_1 \dots \mu_p} = \epsilon^{\mu_1 \dots \mu_p} \partial_{\mu_1} X^{\mu_1} \dots \partial_{\mu_p} X^{\mu_p}$$

$\rightarrow$ Automatically conserved

Charge:
$$Z^{\mu_1 \dots \mu_p} \equiv q \int d^p \xi j^{\mu_1 \dots \mu_p}$$

• We can recast the statement p-brane couples to a (p+1)-form field as:

$\hookrightarrow$ (p+1)-background gauge field couples to a current

$$\int d^{p+1} \xi j^{\mu_1 \dots \mu_p} \partial_{\mu_1} X^{\mu_1} \dots \partial_{\mu_p} X^{\mu_p} A_{\mu_1 \dots \mu_p}$$

• Note:

$$\begin{aligned} Z^{\mu_1 \dots \mu_p} &= q \int d^p \xi \epsilon^{\mu_1 \dots \mu_p} \partial_{\mu_1} X^{\mu_1} \dots \partial_{\mu_p} X^{\mu_p} \\ &= q \int d^p \xi \partial_{\mu_1} (\epsilon^{\mu_1 \dots \mu_p} X^{\mu_1} \partial_{\mu_2} X^{\mu_2} \dots) \end{aligned}$$

• For a brane wrapped on a p-cycle on M

$q^{-1} Z^{\mu_1 \dots \mu_p}$: measures the # of times the p-brane wraps around the p-cycle.

* $Z^{\mu_1 \dots \mu_p}$: Central Charge of Susy

• For p-branes in a theory $\Rightarrow Z^{(p)}$ central charges in the theory from current

Simple Superbranes: Dynamics completely described by the embedding

$Z^N(x^\alpha, \theta^\alpha)$ coordinates

$D=11$: M2 brane

$S = S_1 + S_2$

$D=10$: F1-string

$g_{mn} = \partial_m Z^N \partial_n Z^M g_{NM}$

$$S_1 = -T \int d^{p+1} \xi \sqrt{-\det(g_{mn})}$$

$$g_{mn} = E_N^a E_N^b \eta_{ab}$$

$$S_2 = -T \int B_{(p+1)}$$

$$B_{(p+1)} = dz^{p+1} \wedge \dots \wedge dz^{N_1} B_{N_1 \dots N_{p+1}} = \frac{1}{(p+1)!} E^{A_1} \wedge \dots \wedge E^{A_{p+1}} B_{A_1 \dots A_{p+1}}$$

$$\text{where } E^A = dz^N E_N^A$$

$$S_2 = -\frac{T}{(p+1)!} \int d^{p+1} \xi \epsilon^{n_1 \dots n_{p+1}} \partial_{n_1} z^{M_1} E_{M_1}^{A_1} \dots \partial_{n_{p+1}} z^{M_{p+1}} E_{M_{p+1}}^{A_{p+1}} B_{A_1 \dots A_{p+1}}$$

• $-T$: interpreted as the charge of the brane which is coupled to the gauge field.

→ Fixed by requiring maximal susy.

• Equality of T from S_1 & S_2 can be thought of as a BPS condition since the charge must be balanced by the mass

$$\Leftrightarrow \textcircled{q = m} \text{ (BPS)}$$

* Geometry of background superspace same ($E_N^a \oplus B_{(p+1)}$)

no torsion T_{mnk} and curvature: has a superfield strength

$$H^{(p+2)} = dB$$

• Now also have:

$$D_{A_1} H_{A_2 \dots A_{p+2}} + T_{A_1 A_2}^B H_{B A_3 \dots A_{p+1}} + (\text{supercyclic perms}) = 0$$

• H_A : superfield strength (superspace tangent space)

$\mathcal{S}$: susy invariant → possesses background susy

$$\delta X^a = \frac{i}{2} \bar{\epsilon} \gamma^a \Theta, \quad \delta \Theta^a = \epsilon^a ?$$

$$E_M^A = \begin{pmatrix} e_{\hat{m}}^a & 0 \\ -\frac{i}{2} \bar{\Theta}^B (\gamma^a)_{B\alpha} & \delta_{\hat{m}}^a \end{pmatrix}$$

$$\text{torsion: } T_{\alpha\beta}^a = i (\gamma^a C^{-1})_{\alpha\beta}$$

$$H_{\alpha\beta a_1 \dots a_p} = -i (-1)^{\frac{1}{4} p(p-1)} (\gamma_{a_1 \dots a_p} C^{-1})_{\alpha\beta}$$

$$T^a = \frac{1}{2} d\bar{\Theta} \wedge (\gamma^a d\Theta)$$

$$H = \frac{i}{2p!} E^{a_p} \wedge \dots \wedge E^{a_1} d\bar{\Theta} \wedge (\gamma_{a_1 \dots a_p} d\Theta)$$

• H is exact, $H = dA$, $dH = 0$

$$\text{yields } d\bar{\Theta} \gamma_a d\Theta d\bar{\Theta} \gamma^{a_{b_1} \dots b_{p-1}} d\Theta = 0$$

• Θ : Grassmann odd

$d\Theta$: Grassmann even

$$\partial \in d\mathbb{R}^M$$

$$F^a_i = E^a_M dz^M$$

$$F^a_i = e^a_M dx^M + \frac{i}{2} \bar{\Theta}^B (\gamma^a)_{B\alpha} d\Theta^\alpha$$

$$\text{Thus, } dF^a_i = d e^a_M \wedge dx^M - \frac{i}{2} d\bar{\Theta}^B (\gamma^a)_{B\alpha} d\Theta^\alpha$$

and all terms vanish separately

$$d\bar{\Theta} \gamma_a d\Theta d\bar{\Theta} \gamma^{a_1 \dots a_{p-1}} d\Theta = 0$$

- Can remove $d\Theta$ so long as we replace it with a grassmann even spinor. (Must include a projector if the spinor is well : $d\Theta \rightarrow (1 + \gamma_*) \chi$ and symmetrise on the spinor indices.

ex: $d\bar{\Theta} = d\Theta^T C$, if Θ is Majorana
 Since $(d\Theta)^T$ is even they factor out.

$$(d\Theta^T)^\alpha (d\Theta)^{\beta_1 \dots \beta_{p-1}} (d\Theta)^\beta (d\Theta)^\epsilon \Big] = 0$$

Since $d\Theta \neq 0$

$$(\gamma_a C^{-1})_{\alpha\beta} (\gamma^{a_1 \dots a_{p-1}} C^{-1})_{\beta\epsilon} = 0$$

$$\begin{aligned} \bar{\chi} \gamma^M \chi &= \bar{\chi}^B (\gamma^M)_B{}^\delta \chi_\delta \\ &= (\chi^T)_\alpha C^{\alpha\beta} (\gamma^M)_B{}^\delta \chi_\delta \\ &= \bar{\chi}^B (\gamma^M)_B{}^\delta C^{-1}_{\delta\gamma} \chi^\gamma \\ &= \bar{\chi}^B (\gamma^M C^{-1})_{B\gamma} \chi^\gamma \end{aligned}$$

* Simple branes can therefore only exist if the above relationship holds (for Majorana)

→ Equal $B = F$ D.O.F from $X^{a'}$, $\Theta^{\alpha'}$ imply a hidden susy on the world volume

* Note: we found

$$T_{a\bar{b}}^a = i(\gamma^a C^{-1})_{\alpha\beta}$$

$$H_{a\bar{b}a_1 \dots a_p} = -i(-1)^{\frac{1}{2}p(p-1)} (\gamma_{a_1 \dots a_p} C^{-1})_{\alpha\beta}$$

which together give us exactly the above condition when put together.

K-symmetry:

$$\delta_{\kappa} Z^M E_M^{\alpha} = 0$$

$$\delta_{\kappa} Z^M E_M^{\alpha} = (1 + \Gamma)^{\alpha}_{\beta} \kappa^{\beta}$$

$\kappa^{\beta}(\xi)$: local parameter (world volume scalar)
(space time spinor)

$$\Gamma = \frac{(-1)^{\frac{1}{2}(p-2)(p-1)}}{(p+1)! \sqrt{-\det(g_{mn})}} \epsilon^{n_1 \dots n_{p+1}} \partial_{n_1} Z^{M_1} E_{M_1}^{\alpha_1} \dots \partial_{n_{p+1}} Z^{M_{p+1}} E_{M_{p+1}}^{\alpha_{p+1}} \gamma_{\alpha_1 \dots \alpha_{p+1}}$$

$\Gamma^2 = 1$, $\frac{1}{2}(1 + \Gamma)$: projectors

• If $\kappa = \frac{1}{2}(1 - \Gamma)\tilde{\kappa}$, then there is no contribution from the symmetry $\Rightarrow$ only half of the components of κ contribute $\Rightarrow$ Gauges $16 \ominus \rightarrow 8 \ominus$

• For a flat background super space $g_{MN} = \eta_{MN}$:
 $\delta X^{\mu} = \frac{i}{2} \Theta \gamma^{\mu} \delta \Theta$, $\delta \Theta = (1 + \Gamma)\kappa$

└────────────────────────────────┘
G.S. superstring

D-branes: have a vector field A_n living on their worldvolume along with $(X^{\mu}, \Theta^{\alpha}, A_n)$

• Action is a generalized version of that of the simple brane to include A_n .

Born-Infeld Action:

$$-T \int d\xi^p \sqrt{-\det(g_{mn} + c F_{mn})}$$

$F_{mn} = 2 \partial_{[m} A_{n]}$, D.O.F of A_n : $(p-1)$

* let's look at $p=1$: $-T \int d^2 \xi \sqrt{-\det(g_{mn} + c F_{mn})}$

However Born & Infeld started with the action

$$S_{BI} \sim \int \sqrt{-\det(\eta_{\alpha\beta} + \kappa F_{\alpha\beta})} d^4 \sigma$$

* Take $\Sigma \cong \mathbb{R} \times S^1$

$$G_{\alpha\beta} = \eta_{\alpha\beta} + \kappa F_{\alpha\beta} = \begin{pmatrix} -1 & \kappa F_{01} \\ \kappa F_{01} & 1 \end{pmatrix}$$

$$\det(G_{\alpha\beta}) = -1 + \kappa^2 F_{01}^2$$

$$\therefore -\det(G_{\alpha\beta}) = 1 - \kappa^2 F_{01}^2$$

$$\leadsto \int d^2 \xi \sqrt{1 - \kappa^2 F_{01}^2} \quad (\text{string})$$

• By T-duality should be dual to D0-brane on a dual circle
 $\frac{1}{\kappa} \sim \frac{1}{\sqrt{\alpha'}} \sim \frac{1}{\ell_s}$

$A_1 = \frac{1}{2\pi\alpha'} X$ X : coordinate on dual circle

$$F_{01} = 2 \partial_0 A_1 = 2 \left(-\frac{1}{2\pi\alpha'}\right) \partial_0 \tilde{X}_1 = \frac{-1}{2\pi\alpha'} \dot{\tilde{X}}^1$$

$$V \equiv \dot{\tilde{X}}^2$$

$$F_{01} = \frac{-1}{2\pi\alpha'} V, \quad K \equiv 2\pi\alpha'$$

$$-T \int d^2\sigma \sqrt{1 - v^2 F_{01}^2} = -T 2\pi R' \int d\tau \sqrt{1 - v^2}$$

$$\Rightarrow m = T 2\pi R' = \frac{R'}{\alpha'}, \quad m = \frac{R'}{\alpha'} \quad \left(\begin{array}{l} \text{fine tune } R' \text{ for} \\ \text{mass} \end{array} \right)$$

$$(BI \text{ string}) \xleftrightarrow{T\text{-Duality}} (\text{Particle}) \quad (\text{D0-brane Action})$$

* BI is required for Lorentz invariance of T-dual theory

$$\text{General: } S = -T \int d^{p+1} \xi \sqrt{-\det(g_{mn} + c F_{mn})}$$

$$g_{mn} = \partial_m Z^\mu \partial_n Z^\mu g_{\mu\nu}$$

* Expand:

$$\sqrt{-\det(g_{np} + c F_{np})} = \sqrt{-\det(g_{nm})} \sqrt{\det(\delta_p^m + c F_p^m)}$$

$$\left(\det(\delta_p^m + c F_p^m) \right)^{1/2} = e^{\frac{1}{2} \text{tr}(\ln(\delta_p^m + c F_p^m))}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

→ all odd terms vanish

$$= \sqrt{-|g_{mn}|} \left[\frac{1}{2} \text{Tr} \left(-\frac{1}{2} c^2 F_l^m F_p^l - \frac{c^4}{4} F_l^m F_q^l F_r^q F_p^r \right) \right]$$

$$= \sqrt{-|g_{mn}|} \left(1 + \frac{c^2}{4} F_{mn} F^{mn} - \frac{c^4}{8} F_p^m F_q^p F_r^q F_n^r + \frac{c^4}{32} (F_{mn} F^{mn})^2 + \dots \right)$$

$$\text{Oh: } \text{tr}(F_l^m F_p^l) = -F_{ml} F^{ml}$$

$$F_l^m F_p^l = F^{ml} F_{em} = -F^{ml} F_{me} \quad \checkmark$$

* Note: we get Maxwell and gravity + higher order terms

$$\frac{T_p c^2}{4} = \frac{1}{4g^2} \Leftrightarrow c^2 = \frac{1}{g^2 T_p} \Rightarrow c^2 = \frac{2}{T_p g^2}$$

(These are for a Bosonic theory)

* Add susy:

$$S_1 = -T_p \int d^p \xi e^{-\phi} \sqrt{-\det(g_{mn} + F_{mn}^2)}$$

ϕ : dilaton

{Roll-back of

$$F_{mn} = 2\pi\alpha' F_{mn} + B_{nm}$$

$$B_{nm} = \frac{\partial x^n}{\partial \xi^n} \frac{\partial x^m}{\partial \xi^m} B_{nm} \quad \left\{ \begin{array}{l} \text{Background} \\ B \text{ to } M \end{array} \right.$$

- S_1 : involves all the massless fields from NS&NS sector of string

$$\text{NS}\otimes\text{NS}: (\phi, g_{mn}, B_{mn})$$

$$[0000]_8 \quad [2\alpha\alpha]_8 \quad [0100]_8^{NS}$$

- On D_p : $\delta A_n = \partial_n \Lambda$ for F (all others next?)

- On M : δF is invariant under (pull back)

$$\delta B_{mn} = 2\partial_{[n} \Lambda_{m]} \quad , \quad \delta A_n = -\frac{1}{2\alpha'} \Lambda_m \quad \Big| \quad \Lambda_n = \partial_n x^m \Lambda_m$$

i.e. the pull back from M to D_p .

- From susy, $B = (\bar{\Theta}^2 \Gamma_{\mu\nu} d\Theta^2 - \bar{\Theta}^2 \Gamma_{\mu\nu} d\Theta^2) \wedge (dx^\mu - \frac{1}{2} \bar{\Theta}^A \Gamma^\mu d\Theta^A)$

* In order to ensure we have K -symmetry we must add a C.S. term (Noether Method)

$$S_2 = g \int \{ C \wedge e^\mu \}_{p+1} \wedge \mathcal{G}$$

C constructed from RR fields pulled back onto D_p including all of its dual fields. Formal sum $C \in \Lambda^\bullet(TD_p)$

$$C = \dots + C_{n_1 \dots n_q} d\xi^{n_1} \wedge \dots \wedge d\xi^{n_q} + \dots$$

$$= \dots + C_{n_1 \dots n_q} \frac{\partial x^{n_1}}{\partial \xi^{n_1}} \dots \frac{\partial x^{n_q}}{\partial \xi^{n_q}} d\xi^{n_1} \wedge \dots \wedge d\xi^{n_q} + \dots$$

$$\text{In } \text{IIA}: C \in \bigoplus_n \Lambda^{2n+1}(TD_p) \text{ (odd)}$$

$$\text{IIB}: C \in \bigoplus_n \Lambda^{2n}(TD_p) \text{ (even)}$$

$$D_p = \chi^*(T M)$$

$$\text{Susy: } \begin{cases} \delta_\epsilon \Theta = \epsilon \\ \delta_\epsilon x^m = \epsilon \Gamma^m \Theta \end{cases}$$

$$e^\mu = \exp(F_{mn} d\xi^n \wedge d\xi^m) = 1 + F_{mn} d\xi^m \wedge d\xi^n + \dots$$

$\{ \cdot \}_{p+1}$: Means take the $(p+1)$ -form from the expansion

$\mathcal{G} = 1 + \dots$, $\dots$ comes from the Riemann Curvature.

- D1 string in IIB (F is simple)

$$S_2 = g \int \{ C \wedge e^\mu \}_2 \wedge \mathcal{G}$$

$$C = C^{(0)} + C^{(2)} + \dots$$

$$= g \int C \wedge (1 + F^{(2)} + F^{(4)} + \dots) \wedge \mathcal{G} \quad \leftarrow \text{ignore}$$

$$= g \int (C^{(0)} + C^{(2)}) \wedge (1 + F^{(2)} + \dots)$$

$$= g \int (C^{(2)} + C^{(0)} F^{(2)})$$

$$S_2 = g \int (C^{(0)} F_{mn} + C_{nm}) d\xi^n \wedge d\xi^m \quad (\text{D1-brane})$$

* N -coincident branes possess vectors that belong to rep of $U(N)$ -gauge theory

↳ lowest order is just that of 10D $\mathcal{N}=1$ dm reduced to (p+1).

$$S_{\text{Bosonic}} = \int d^4x \text{Tr} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \sum_i D_\mu \phi_i D^\mu \phi_i - \frac{1}{4} \left(\sum_{ij} [\phi_i, \phi_j]^2 \right) \right)$$

↳ Minimum: $\phi_i \in \mathcal{H}$ (Cartan-sub alg)

Can be thought of as the positions of the N -D_p's